



VICTORIA JUNIOR COLLEGE
JC 1 PROMOTIONAL EXAMINATIONS
Higher 2

CHEMISTRY

9729//01

Paper 1 Multiple Choice

2 October 2020

1 hour

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your index number, name and CT group on the Answer Sheet.

There are **thirty** questions. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of **13** printed pages.

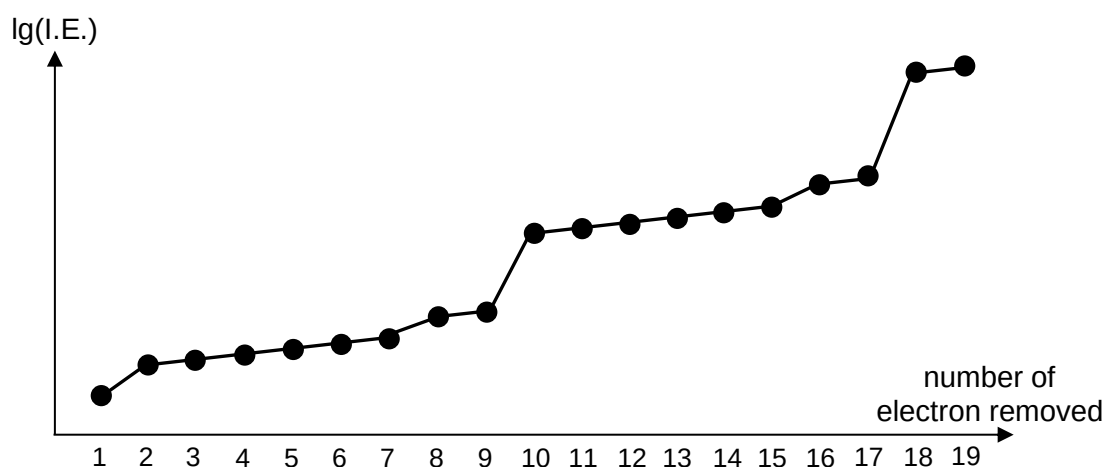
For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the **one** you consider to be correct.

- 1 In a particular experiment, a beam of ${}^9\text{Be}^{2+}$ ions is deflected by an electric field through an angle of $+10^\circ$.

Which one of the following particles, when passed through the same electric field, will be deflected through an angle half the magnitude of ${}^9\text{Be}^{2+}$?

	proton number	neutron number	electronic configuration
A	6	8	$1s^2 2s^2 2p^6$
B	8	10	$1s^2 2s^2 2p^6$
C	12	8	$1s^2 2s^2 2p^6$
D	15	20	$1s^2 2s^2 2p^6$

- 2 The graph below shows the logarithm of the ionization energies for all 19 electrons of Z^{3+} , where **Z** represents an element in the Periodic Table.



Which statement about element **Z** is correct?

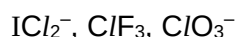
- A** Element **Z** is in Group 1.
- B** Element **Z** is in Period 3.
- C** Element **Z** is a d-block element.
- D** Element **Z** has no half-filled orbitals.

- 3 SiF_4 reacts with F^- to produce the hexafluorosilicate ion, SiF_6^{2-} but CF_4 does not.

Which statement about the above phenomenon is **false**?

- A 6 sigma bonds exists in SiF_6^{2-} .
- B F^- bonds to SiF_4 via ionic bonds.
- C SiF_6^{2-} has an octahedral arrangement about the central Si atom.
- D CF_4 behaves differently because carbon does not have energetically available vacant d orbitals to expand its octet configuration.

- 4 Consider the following chlorine-containing species.



Which statements about these species are correct?

- 1 The bond angles increase in the order of ClF_3 , ClO_3^- , ICl_2^- .
- 2 Two of the three species have an overall dipole moment.
- 3 Only ClO_3^- is not planar.

- A 1, 2 and 3
- B 1 and 2 only
- C 2 and 3 only
- D 1 only

- 5 Information about some hydrocarbons are given below.

Hydrocarbon	boiling point / °C
ethane, CH_3CH_3	-89.0
propane, $\text{CH}_3\text{CH}_2\text{CH}_3$	-42.0
butane, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	-1.7
2-methylpropane, $(\text{CH}_3)_3\text{CH}$	-11.7

Which statements help to explain the variation in boiling points of the hydrocarbons?

- 1 Presence of more hydrogen atoms results in stronger hydrogen bonding.
- 2 Presence of more covalent bonds results in more energy required to break them.
- 3 Greater surface area of contact results in stronger instantaneous dipole-induced dipole interactions.
- 4 Presence of more electrons results in stronger instantaneous dipole-induced dipole interactions.

- A 1, 2 and 3 only
- B 2 and 3 only
- C 3 and 4 only
- D 4 only

- 6 Cations with high charge density can polarise adjacent anions. This results in ionic bonds with covalent character. If the polarisation is extensive enough, the bond formed might be classified as a covalent bond instead.

Consider the various fluorides and chlorides formed by the Group 14 metals, Sn and Pb.

Which option lists the most ionic and the most covalent halide compounds?

	most ionic	most covalent
A	SnF_2	PbCl_4
B	PbF_2	SnCl_4
C	SnF_4	PbCl_2
D	PbF_4	SnCl_2

- 7 Antimony, Sb, can be produced via a two-stage process from its ore stibnite, Sb_2S_3 .

The ore is first roasted in oxygen, producing Sb_4O_6 and SO_2 . The Sb_4O_6 is then reduced by carbon, producing Sb and CO_2 .

What volume of CO_2 , measured under standard temperature and pressure, will be produced when 5 moles of Sb_2S_3 are processed?

- A 57 dm³ B 170 dm³ C 227 dm³ D 341 dm³

- 8 Diamond is a pure form of carbon. The mass of a diamond can be measured in carats, where one carat is equivalent to 0.200 g of carbon.

Which expression gives the number of carats in a diamond that contains 6.02×10^{23} carbon atoms?

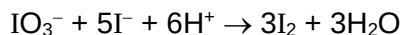
- A 0.200×12.0 C $\frac{12.0}{0.200}$
- B $\frac{0.200}{12.0}$ D $\frac{0.200}{6.02 \times 10^{23}} \times 12.0$

- 9 SO_2 gas is found in air near volcanoes. 201.6 cm³ of SO_2 collected at s.t.p. near a volcano required 25.0 cm³ of 0.120 mol dm⁻³ acidified potassium dichromate(VI) solution for complete reaction.

What is the final oxidation state of sulfur in the product mixture?

- A 0 B +2 C +4 D +6

- 10 Iodate(V) ions react with iodide ions under acidic conditions as follows.



The acid is added to an aqueous solution containing excess iodate(V) ions and iodide ions. The liberated iodine is then titrated with sodium thiosulfate solution.

22.00 cm³ of 0.100 mol dm⁻³ sodium thiosulfate solution was required to react with the iodine produced when 25.0 cm³ of sulfuric acid were used.

What was the concentration of the sulfuric acid in mol dm⁻³?

- | | | | |
|----------|--------|----------|--------|
| A | 0.0440 | C | 0.0587 |
| B | 0.0468 | D | 0.0880 |
- 11 Given that the enthalpy change of solution of ammonium nitrate is +25.7 kJ mol⁻¹.

What is the expected change in temperature upon dissolving 2.50 g of ammonium nitrate crystals into 25.0 cm³ of water in a styrofoam cup?

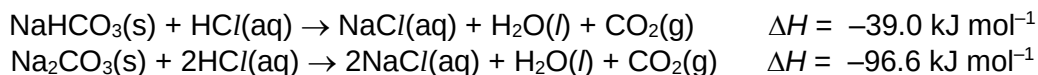
[Density of water = 1.00 g cm⁻³, specific heat capacity of water = 4.2 J g⁻¹ K⁻¹]

- | | | | |
|----------|--|----------|--|
| A | $\frac{2.50 \times 25.7 \times 10^3}{80.0 \times 25.0 \times 4.2} ^\circ\text{C}$ | C | $\frac{80.0 \times 25.7 \times 10^3}{2.50 \times 25.0 \times 4.2} ^\circ\text{C}$ |
| B | $\frac{2.50 \times (-25.7 \times 10^3)}{80.0 \times 25.0 \times 4.2} ^\circ\text{C}$ | D | $\frac{80.0 \times (-25.7 \times 10^3)}{2.50 \times 25.0 \times 4.2} ^\circ\text{C}$ |
- 12 Which expression gives the same value as the standard enthalpy change of formation of carbon monoxide?

[Given that $\Delta H^\circ_c(\text{X})$ = standard enthalpy change of combustion of X; $\Delta H^\circ_f(\text{X})$ = standard enthalpy change of formation of X]

- A** $\Delta H^\circ_f(\text{CO}_2) - \frac{1}{2} \Delta H^\circ_c(\text{C}(\text{graphite}))$
- B** $\Delta H^\circ_f(\text{CO}_2) - \Delta H^\circ_c(\text{C}(\text{graphite}))$
- C** $\Delta H^\circ_f(\text{CO}_2) - \Delta H^\circ_c(\text{CO})$
- D** $\frac{1}{2} \Delta H^\circ_c(\text{C}(\text{graphite}))$

- 13** The following data are needed for this question.



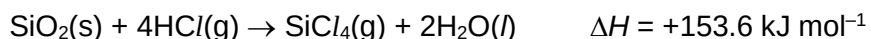
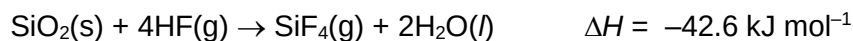
On heating, sodium hydrogencarbonate decomposes as shown.



What is the enthalpy change for this decomposition?

- | | | | |
|----------|-----------------------------|----------|-----------------------------|
| A | $-57.6 \text{ kJ mol}^{-1}$ | C | $+18.6 \text{ kJ mol}^{-1}$ |
| B | $-18.6 \text{ kJ mol}^{-1}$ | D | $+57.6 \text{ kJ mol}^{-1}$ |
- 14** One of the components of glass is SiO_2 . The reaction between SiO_2 and HF is spontaneous at room temperature. However, the reaction between SiO_2 and HCl is not spontaneous at room temperature.

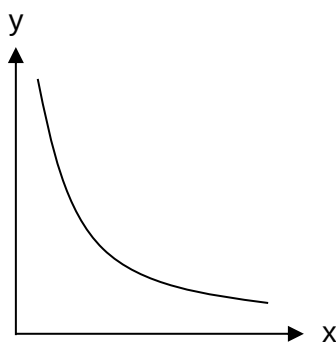
The reactions of interest are given below.



Which of the following statements are correct?

- | | |
|----------|--|
| 1 | The entropy change for both reactions are negative at room temperature. |
| 2 | The entropy change for both reactions are positive above 100°C . |
| 3 | The magnitude of $T\Delta S$ is smaller than that of ΔH for the reaction with HF at room temperature. |
- | | | | |
|----------|----------------------------|----------|----------------------------|
| A | 1 only | C | 1 and 3 only |
| B | 2 and 3 only | D | 1, 2 and 3 |

- 15** Which options correctly represent the ideal gas behaviour as shown by the graph below?



	y-axis	x-axis	variables held constant
1	V	T	n and p
2	V	$\frac{1}{p}$	n and T
3	p	V	n and T
4	p	$\frac{1}{T}$	n and V

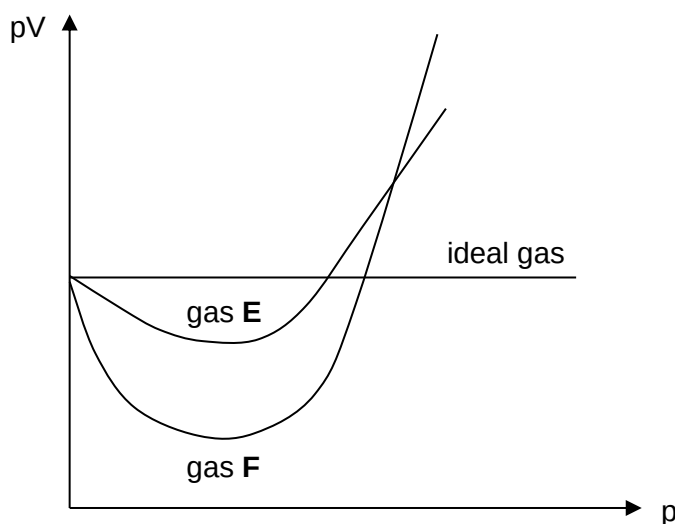
- A** **1 and 2 only**
- B** **1 and 3 only**
- C** **2 and 4 only**
- D** **3 and 4 only**

- 16** A 5.0 dm^3 sample of oxygen at a pressure of 500 kPa and 3.0 dm^3 sample of nitrogen at a pressure of 300 kPa are both introduced into a 2.0 dm^3 vessel at room temperature.

Assuming the two gases do not react with each other, what is the total pressure in the vessel at room temperature?

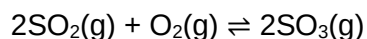
- | | | | |
|----------|----------|----------|----------|
| A | 750 kPa | C | 1600 kPa |
| B | 1200 kPa | D | 1700 kPa |

- 17 The graphs below show the deviation of two gases, **E** and **F**, from ideality.



Which statement about the graphs is true?

- A Gas **E** could be CH_4 while gas **F** could be CO_2 .
 - B Gas **E** could be NH_3 while gas **F** could be H_2 .
 - C Gas **E** could have a larger relative molecular mass than gas **F**.
 - D Gas **E** could form stronger intermolecular instantaneous dipole-induced dipole attractions than gas **F**.
- 18 SO_2 reacts with O_2 according to the following equation.

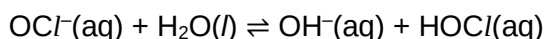


1 mol of SO_2 and 1 mol of O_2 are added into a closed vessel and left to react. After equilibrium has been reached, the partial pressure of SO_3 is found to be 50% of the total pressure.

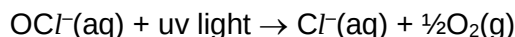
What is the mole fraction of SO_2 in the equilibrium mixture?

- A $\frac{1}{8}$ B $\frac{1}{4}$ C $\frac{1}{3}$ D $\frac{1}{2}$

- 19 Swimming pool water can be kept free of harmful bacteria by adding aqueous sodium chlorate(I), NaOCl. This reacts with water to produce HOCl molecules which kill bacteria.



In bright sunlight, the OCl^- ion is broken down by UV light.

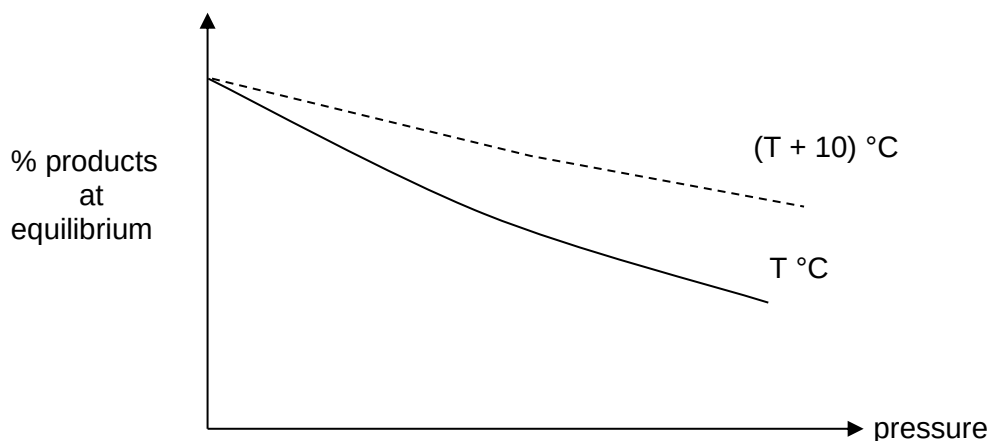


Which method would increase the concentration of $\text{HOCl}(\text{aq})$?

- 1 Build an indoor swimming pool to reduce exposure to sunlight.
- 2 Acidify the pool water.
- 3 Add a solution of chloride ions.
- 4 Add a solution of hydroxide ions.

- | | |
|-----------------------|--------------------------|
| A 1 and 2 only | C 1 and 3 only |
| B 2 and 3 only | D 2, 3 and 4 only |

- 20 The graphs below show the variation of the percentage of gaseous products present at equilibrium, with temperature and pressure.



Which system could the graphs represent?

- | | |
|--|--|
| A $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$ | $\Delta H = -92 \text{ kJ mol}^{-1}$ |
| B $3\text{O}_2(\text{g}) + 4\text{NH}_3(\text{g}) \rightleftharpoons 2\text{N}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$ | $\Delta H = -1248 \text{ kJ mol}^{-1}$ |
| C $2\text{N}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{N}_2\text{O}(\text{g})$ | $\Delta H = +82 \text{ kJ mol}^{-1}$ |
| D $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightleftharpoons 2\text{CO}(\text{g})$ | $\Delta H = +173 \text{ kJ mol}^{-1}$ |

- 21** When 0.010 mol of a weak acid HA is dissolved in 1 dm³ distilled water, the pH of the resultant solution is 4.6.

What is the numerical value of the acid dissociation constant, K_a of this weak acid HA?

- | | | | |
|----------|-----------------------|----------|-----------------------|
| A | 1.6×10^{-11} | C | 6.3×10^{-10} |
| B | 2.5×10^{-10} | D | 6.3×10^{-8} |

- 22** Which of the following pairs of solutions would form an acid buffer when mixed together?

- 1 HCN and KCN
- 2 HNO_3 and KNO_3
- 3 KOH and KCl

- A** 1 only
- B** 2 only
- C** 1 and 2 only
- D** 2 and 3 only

- 23** What is the pH of the resultant solution when 100 cm³ of 0.100 mol dm⁻³ aqueous NH₃ and 80 cm³ of 0.150 mol dm⁻³ aqueous NH₄Cl are mixed at 25 °C? (pK_b of NH₃ = 4.75)

- A** 8.2 **B** 9.2 **C** 10.2 **D** 11.2

- 24** The pH range and colour changes of two acid-base indicators are given below.

indicator	colour in acidic solution	pH range	colour in basic solution
X	violet	3.0 – 5.0	red
Y	yellow	5.7 – 7.6	blue

Which solution will appear red in **X** and yellow in **Y**?

- A** 0.10 mol dm⁻³ of HCl
B 0.10 mol dm⁻³ of NaOH
C 0.10 mol dm⁻³ CH₃CO₂H ($K_a = 1.8 \times 10^{-5}$)
D 0.10 mol dm⁻³ of HA ($K_a = 2.5 \times 10^{-10}$)

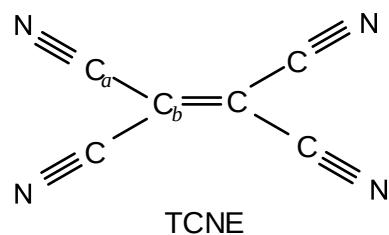
- 25 The solubility products of three calcium salts are shown below.

calcium salt	Ca(OH)_2	CaCO_3	CaF_2
K_{sp}	5.5×10^{-6}	8.7×10^{-9}	4.0×10^{-11}

Calcium nitrate is gradually added to a 1 dm^3 solution containing carbonate, fluoride and hydroxide ions, each with a concentration of $1.0 \times 10^{-3} \text{ mol dm}^{-3}$.

What is the correct order at which these calcium salts precipitate out of solution?

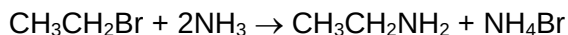
- A Ca(OH)_2 , CaF_2 , CaCO_3
 B Ca(OH)_2 , CaCO_3 , CaF_2
 C CaCO_3 , CaF_2 , Ca(OH)_2
 D CaF_2 , CaCO_3 , Ca(OH)_2
- 26 Tetracyanoethylene (TCNE) is used in the preparation of organic superconductors.



Which set below matches the type of hybridisation to the correct carbon atom in TCNE?

	C_a	C_b
A	sp	sp^2
B	sp^2	sp
C	sp^2	sp^2
D	sp^3	sp^3

- 27 An amine is produced in the following reaction.

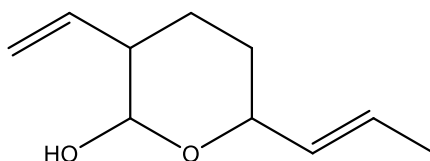


Which of the following statements about the reaction are true?

- 1 NH_3 behaves as an electrophile.
- 2 It follows a nucleophilic substitution mechanism.
- 3 NH_4Br is produced via an acid-base reaction.

- | | |
|----------------|----------|
| A 1 and 3 only | C 1 only |
| B 2 and 3 only | D 3 only |

- 28 Which statements about the following molecule are correct?



- 1 It is unsaturated.
 - 2 It exhibits cis-trans isomerism.
 - 3 It contains an alcohol functional group.
- | | |
|----------------|----------------|
| A 1, 2 and 3 | C 2 and 3 only |
| B 1 and 3 only | D 2 only |

- 29 Ethane reacts with chlorine gas in the presence of UV light to form a mixture of products.

Which statement regarding the reaction is **false**?

- A Both initiation and propagation steps produce chlorine radicals.
- B Both propagation and termination steps produce hydrogen chloride.
- C Both propagation and termination steps involve C–Cl bond formation.
- D Both initiation and propagation steps involve homolytic fission.

- 30** Chlorofluoroalkanes are commonly used as aerosol propellents. However, they cause ozone depletion when they rise to the stratosphere. It has been suggested that fluoroalkanes should be used instead.

What could be a possible reason for the suggestion?

- A** Fluoroalkanes are less volatile than chlorofluoroalkanes and thus more difficult to reach the stratosphere.
- B** Fluorine radicals may be produced, but unlike chlorine radicals, are not regenerated after reaction with ozone.
- C** Fluorine radicals may be produced, but unlike chlorine radicals, do not react with ozone.
- D** Fluorine radicals are not produced as C-F bonds are stronger than C-Cl bonds.

VICTORIA JUNIOR COLLEGE
2020 JC1 PROMOTIONAL EXAMINATIONS
H2 CHEMISTRY PAPER 1 ANSWERS

1	B	6	B	11	B	16	D	21	D	26	A
2	C	7	B	12	C	17	A	22	A	27	B
3	B	8	C	13	C	18	A	23	B	28	A
4	A	9	D	14	C	19	A	24	D	29	B
5	C	10	A	15	D	20	D	25	C	30	D



VICTORIA JUNIOR COLLEGE
JC 1 PROMOTIONAL EXAMINATIONS
Higher 2

CANDIDATE
NAME

CT GROUP

CHEMISTRY

9729/02

Paper 2 Structured and Free Response

2 October 2020

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your index number, name and CT group on all the work you hand in.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use paper clips, glue or correction fluid.

Section A Structured

Answer **all** questions in the spaces provided on the Question Paper.

Section B Free Response

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
S1	
S2	
E1	
E2	
Total	

This document consists of **18** printed pages and **2** blank pages.

Section A: Structured Questions [30 marks]

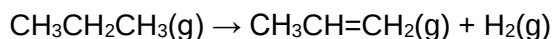
Answer **all** the questions in the spaces provided.

- S1** Many industries employ use of gases or manufacture them in large quantities. This question pertains to a few gases and their properties.

The data for the following gases are shown below, which is needed for relevant parts of this question. For comparison, CH₃OH, a liquid at standard conditions is also included.

Species	M_r	Boiling point / °C	$\Delta H_c^\ominus$ / kJ mol ⁻¹	Density / g cm ⁻³
CH ₃ CH=CH ₂	42.0	-47.6	-2058	1.74 x 10 ⁻³
CO	28.0	-191.5	-283	1.14 x 10 ⁻³
H ₂	2.0	-252.9	-286	8.20 x 10 ⁻⁵
CH ₃ OH	32.0	64.7	-715	0.792

- (a) Propene, CH₃CH=CH₂ can be synthesised via a direct dehydrogenation reaction of propane, CH₃CH₂CH₃, in the presence of a suitable catalyst. The equation is



- (i) Using relevant data from the *Data Booklet*, calculate the enthalpy change of the above reaction.

[2]

- (ii) Explain the sign of ΔS for the above reaction.

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[2]

- (iii) Hence using your answer to (a)(ii), explain how the spontaneity of the reaction will vary as temperature increases.

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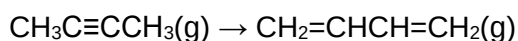
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[2]

- (b) Propene, like many other alkanes and alkenes can also be manufactured through cracking of a hydrocarbon with a higher M_r , for example butadiene, $\text{CH}_2=\text{CHCH}=\text{CH}_2$.

Given the following thermochemical data, and using additional data from the table above, construct an energy cycle and calculate the standard enthalpy change of the following reaction.



Standard enthalpy change of combustion of $\text{CH}_3\text{C}\equiv\text{CCH}_3(\text{g})$	$-2577 \text{ kJ mol}^{-1}$
Standard enthalpy change of combustion of carbon	-394 kJ mol^{-1}
Standard enthalpy change of formation of $\text{CH}_2=\text{CHCH}=\text{CH}_2(\text{g})$	$+110 \text{ kJ mol}^{-1}$

[4]

- (c) Methanol can be manufactured cheaply and efficiently using hydrogen and carbon monoxide.

The graph below shows the behaviour of 1 mol of an ideal gas at a fixed temperature. On the same axes, sketch graphs of propene, hydrogen and carbon monoxide. You may assume that the amount of each gas is 1 mol and at the same temperature as that of an ideal gas. Label each of the graphs clearly.



[2]

- (d) A method of measuring fuel content is to determine a quantity known as energy density, which is the heat released on burning 1 m^3 of the fuel under standard conditions. Thus the units of energy density are in J m^{-3} .
- (i) Calculate the energy density, in J m^{-3} for each of the **organic** species listed in the table above at the beginning of the question. Hence state which species has a higher fuel content.

Higher fuel content:

[3]

- (ii) With reference to the data in the table above, suggest **one** additional advantage of using the species you have indicated in (d)(i) as a fuel source.

.....

.....

.....

[1]

[Total: 16]

S2 The following question involves borane, a highly reactive molecule with the formula BH_3 .

- (a) Draw the dot-and-cross diagram of BH_3 . Use your diagram to suggest why borane is a highly reactive molecule.

.....,

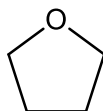
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[2]

- (b) To achieve a more stable structure, borane forms many adducts. An example is $\text{BH}_3 \cdot \text{THF}$. The structure of THF is given below.



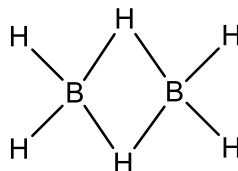
THF

Draw the structure of the $\text{BH}_3 \cdot \text{THF}$ adduct, stating clearly the bond angles about the central atoms involved in the formation of the adduct.

[2]

- (c) Borane dimerises to form diborane, B_2H_6 . The synthesis of diborane in the 19th century baffled scientists. It was not until the 1950s when its molecular structure became better known, with William Nunn Lipscomb, Jr. confirming its structure using X-ray crystallography and developing theories to explain its unique bond arrangement.

- (i) The structure of diborane is shown below.



diborane

Explain what is unique about its structure.

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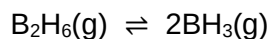
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..... [1]

- (ii) State the shape about each boron atom in diborane.

..... [1]

- (d) In a closed system, diborane and borane exist in equilibrium:



The equilibrium constant value for the equation above between 353 K and 423 K is given by the following expression, where K_p is in Pa and T is in K:

$$\log_{10} K_p = 7.478 - \frac{6205}{T} \dots\dots(1)$$

- (i) Write the K_p expression for the equilibrium reaction between borane and diborane.

[1]

- (ii) Using expression (1), calculate the value of K_p at 353 K and 423 K respectively.

353 K:

423 K:

[1]

- (iii) By considering the bonds involved in diborane and borane, state whether the forward reaction is likely to be exothermic or endothermic. Explain how the values calculated in (d)(ii) are able to confirm this.

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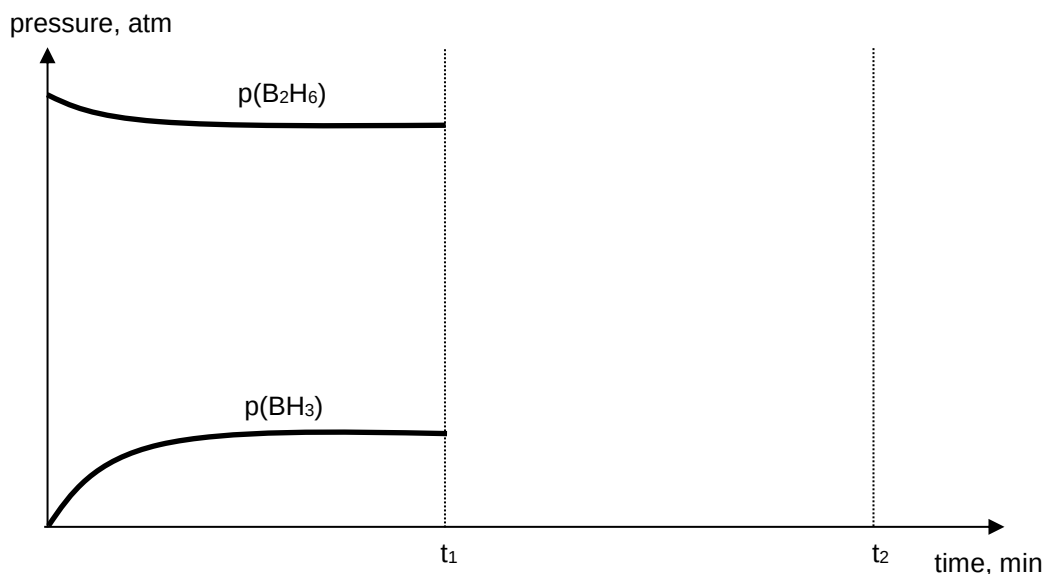
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[3]

The following graphs show the change in partial pressure of diborane and borane with time when a known amount of diborane is heated in a closed system.



The system is said to be in *dynamic equilibrium* at t_1 min.

(iv) Define the term *dynamic equilibrium*.

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[1]

(v) Sketch and label on the diagram above what happens to the partial pressures of diborane and borane from t_1 min to t_2 min when volume of the container suddenly increases at t_1 min at constant temperature.

[Note: t_2 min refers to the time for any new equilibrium to be achieved.]

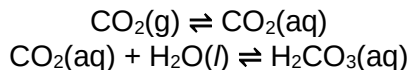
[2]

[Total: 14]

Section B: Essay Questions [40 marks]

Answer **all** the questions in the spaces provided.

- E1** When carbon dioxide gas dissolves in water, carbonic acid (a diprotic acid) with K_{a1} value of $4.3 \times 10^{-7} \text{ mol dm}^{-3}$ is formed. Carbonic acid is a *weak Bronsted-Lowry acid*.



A student was instructed to titrate 25.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ carbonic acid, H_2CO_3 , with $0.0625 \text{ mol dm}^{-3}$ calcium hydroxide, $\text{Ca}(\text{OH})_2$. After the end of the titration, the student noticed that an insoluble white precipitate was formed.

- (a) (i)** What do you understand by the term *weak Bronsted-Lowry acid*.

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[1]

- (ii)** Calculate the pH of $0.100 \text{ mol dm}^{-3} \text{H}_2\text{CO}_3$. Assume that the second dissociation of H_2CO_3 is negligible.

[2]

- (iii)** Calculate the volume of $\text{Ca}(\text{OH})_2$ needed to completely react with 25.0 cm^3 of $0.100 \text{ mol dm}^{-3} \text{H}_2\text{CO}_3$.

[1]

- (iv) Calculate the pH of a solution which contains both H_2CO_3 and HCO_3^- in equal concentrations.

[1]

- (v) The table shows some acid–base indicators.

name of indicator	pH range of colour change
malachite green	0.2–1.8
bromothymol blue	6.0–7.6
thymolphthalein	9.3–10.6

Name a suitable indicator for the above titration carried out by the student. With the aid of an equation, explain your answer.

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[3]

- (b) Human blood is slightly basic with a normal pH of around 7.4. Its pH must be maintained for the effective transportation of oxygen.

The major buffer system which controls the pH of blood is the hydrogencarbonate and carbonic acid buffer system.

Explain, using equations, why an aqueous mixture of HCO_3^- and H_2CO_3 can act as a buffer in blood.

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[2]

- (c) Limewater is a saturated solution of $\text{Ca}(\text{OH})_2$, and is often used to test for the presence of CO_2 formed during experiments. A positive test is indicated by the formation of a white precipitate, CaCO_3 .

- (i) Write an equation, with state symbols, to represent the reaction between limewater and CO_2 .

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[1]

- (ii) The solubility of $\text{Ca}(\text{OH})_2$ is 1.73 g dm^{-3} . Calculate the concentration of Ca^{2+} in mol dm^{-3} , for a saturated $\text{Ca}(\text{OH})_2$ solution.

[1]

- (iii) Calculate the K_{sp} of CaCO_3 given that its solubility is $1.5 \times 10^{-4} \text{ mol dm}^{-3}$.

[1]

- (iv) Hence, calculate the minimum volume of CO_2 at r.t.p., that must be bubbled through 1 cm^3 of limewater in order to give a white precipitate, CaCO_3 .

[3]

- (d) When chlorine is bubbled through a solution of iodine in hot aqueous calcium hydroxide, the two halogens react in the $\text{Cl}_2 : \text{I}_2$ ratio of 7 : 1. A white precipitate, **A**, and a solution of calcium chloride are formed. **A** has the following composition by mass:

Ca, 15.7%; I, 47.4%; H, 1.1%; O, 35.8%

- (i) Calculate the empirical formula of **A**.

[2]

- (ii) Construct a balanced equation for the above reaction.

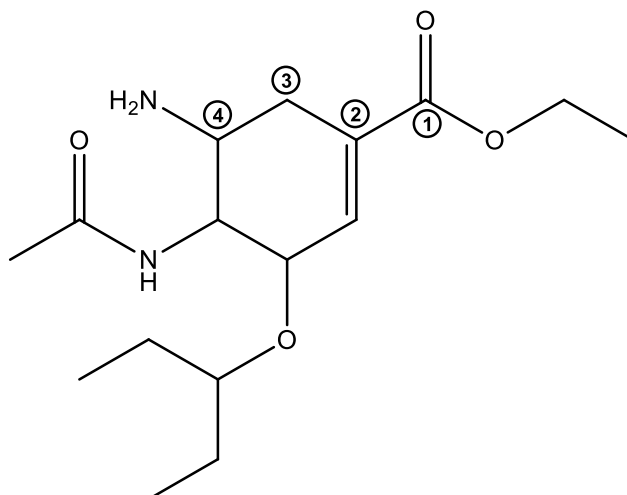
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[1]

- (iii) State the oxidation number of iodine in **A**.

[1]

[Total: 20]

E2 *Tamiflu*[®] is an antiviral medication used for treatment and prevention of the flu.



Tamiflu[®]

- (a) (i) Name any four functional groups that are present in this drug.

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 [2]

- (ii) The bond length of the carbon–carbon single bond between carbon atoms at ① and ② in *Tamiflu*[®] is 0.148 nm while the carbon–carbon single bond between carbon atoms at ③ and ④ is 0.154 nm. Explain the difference in the bond lengths.

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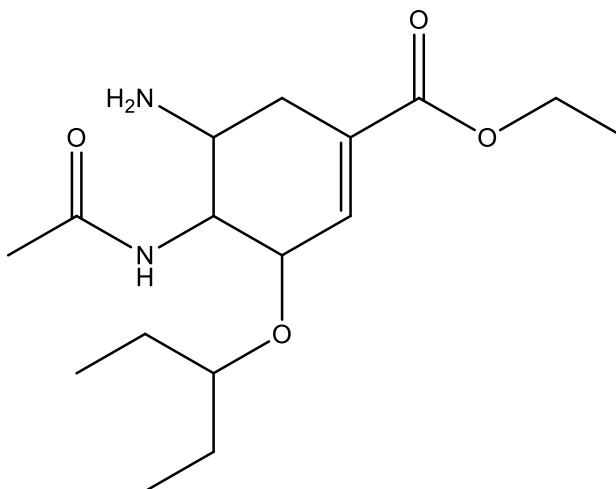
 [2]

- (iii) Does *Tamiflu*[®] display cis-trans isomerism? Explain your answer.

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 [1]

- (iv) On the structure of *Tamiflu*[®] given below, label all the chiral centres with *. Hence determine the total number of stereoisomers of *Tamiflu*[®].



[2]

- (b) A wide range of hydrocarbons can be produced from the refining of crude oil.

- (i) 10 cm³ of hydrocarbon **Z**, C_xH_y in the gaseous state was sparked in an oxygen-rich environment at 150 °C. There was an expansion of 10 cm³. When the gaseous product was treated with sodium hydroxide solution, there was a contraction of 30 cm³.

Deduce the molecular formula of hydrocarbon **Z**.

[2]

- (ii) Write two equations to show the complete and incomplete combustion of 1 mol of the hydrocarbon C_3H_8 . Hence, calculate the total volume of oxygen gas used if 10 cm^3 of C_3H_8 was converted into a mixture of carbon dioxide and carbon monoxide in a mole ratio of 4:1.

[2]

- (c) Both alkanes **A** and **B** have the same molecular formula C_6H_{14} .

- (i) When alkane **B** reacted with chlorine under ultraviolet light, it produced only two isomeric monochlorinated compounds **C** and **D**, with the formula $\text{C}_6\text{H}_{13}\text{Cl}$. Compound **D** was chiral. Suggest the structures of **B**, **C** and **D**.

B	C	D

[2]

- (ii) **A** is a liquid that boils at 68 °C and it has the highest boiling point among its various structural isomers. Identify **A** and explain why it has the highest boiling point in terms of structure and bonding.

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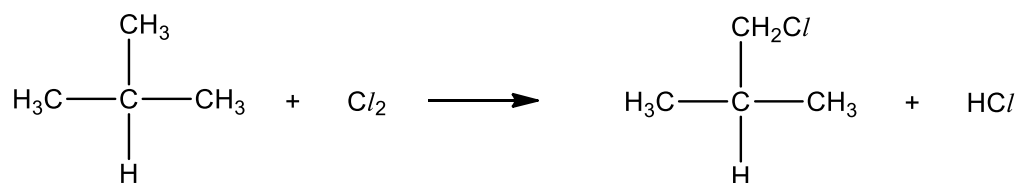
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[2]

- (d) The equation below shows how a chloroalkane can be made by the reaction of the corresponding halogen with an alkane.

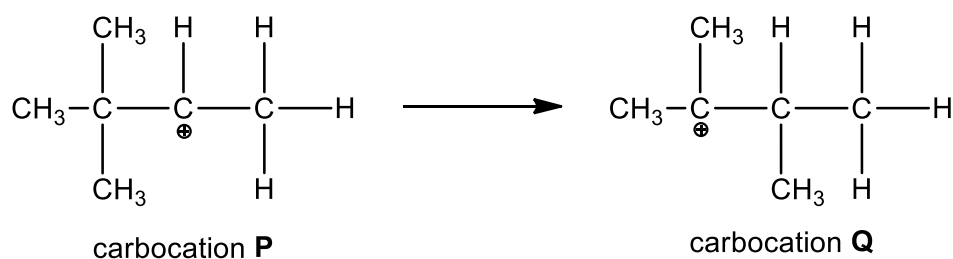


Name and outline the mechanism of this reaction.

[3]

- (e) A 1,2-rearrangement is an organic reaction where a substituent, such as an alkyl group, moves from one atom to another atom in a chemical compound. In a 1,2 shift, the movement involves two adjacent atoms but moves over larger distances are possible.

Below shows an example of the 1,2-rearrangement of the carbocation formed after an alkane undergoes heterolytic fission.



By referring to the relative stability of the carbocations **P** and **Q**, explain why 1,2-rearrangement is a favourable mechanism.

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[2]

[Total: 20]

